

Try 7

Review:

1) At turning point, $y' = 0$

$$y' = 3 - 3x^2$$

$$\therefore x = 1 \text{ or } x = -1$$

$$\text{At } x = -1, y = -2 \checkmark \text{ minimum c)}$$

$$\text{At } x = 1, y = 2 \checkmark \text{ maximum b)}$$

$$2 \quad 2x^2\theta + \tan\theta = 8$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\tan^2\theta + 1 = x^2\theta$$

$$\therefore 2 \tan^2\theta + \tan\theta - 6 = 0$$

$$(2\tan\theta - 3)(\tan\theta + 2) = 0$$

$$\therefore \tan\theta = \frac{3}{2} \rightarrow \theta = 56.3^\circ \checkmark$$

or

$$\tan\theta = -2 \rightarrow \theta = 110.6^\circ$$

$$3 \frac{d}{dx}(\cos x) = -\sin x e^{\cos x} \text{ (c)}$$

4 graph: $\log s$ against t $\log s = (\log a)t + \log k$

$$\text{when } t=0, \log s = 0.3 \therefore \log k = 0.3$$

$$k = 10^{0.3} = 2.0$$

$$\text{gradient} = \log a = 0.4$$

$$a = 6.3 \checkmark$$

$$5 \quad p \begin{pmatrix} 2 \\ -1 \end{pmatrix} + \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 9 \\ 1 \end{pmatrix}$$

$$p = 3 \checkmark$$

$$q = 1 \checkmark$$

Consolidation

1) $\arcsin 0 = 0$ ✓

$\arcsin \frac{1}{2} = \frac{\pi}{6}$ ✓

2 $\arcsin 0.6 = 0.64$ ✓
 $\arccos 0.152 = 1.42$ ✓

3 $\arcsin x = \frac{\pi}{4}$ ✓
 $x = \frac{1}{\sqrt{2}}$ ✓

$\arccos 2x = \frac{\pi}{6}$ ✓
 $x = \frac{\sqrt{3}}{4}$ ✓

$\frac{\pi}{4} - \arctan x = 0$ ✓
 $x = 1$ ✓

4 $\arccos x = 2$ ✓
 $x = \cos 2$ ✓
 $= 0.416$ ✓

$1 - \arcsin x = 0$ ✓
 $x = 0.841$ ✓

$2 + 3 \arccos x = 0$ ✓
 $x = -0.787$ ✓

5 $\arcsin(x-1) = \frac{\pi}{2}$ ✓
 $x-1 = 1$ ✓
 $x = 2$ ✓

6 $\pi - \arccos(2x+1) = -\frac{\pi}{2}$ ✓
 $\arccos(2x+1) = \frac{3\pi}{2} = \frac{\pi}{2}$ ✓
 $2x+1 = 0$ ✓
 $x = -\frac{1}{2}$ ✓

7 $\arctan 4x + \arcsin 1 = \arccos \frac{1}{2}$

$\arctan 4x = \arccos \frac{1}{2} - \arcsin 1$
 $= \frac{1}{3}\pi - \frac{1}{2}\pi$
 $= -\frac{\pi}{6}$
 $4x = \tan -\frac{\pi}{6}$
 $4x = -\frac{\sqrt{3}}{3}$ ✓
 $x = \frac{-\sqrt{3}}{12} = -0.144$ ✓

8 $\arcsin(\sin(-\frac{\pi}{6})) = -\frac{\pi}{6}$ ✓

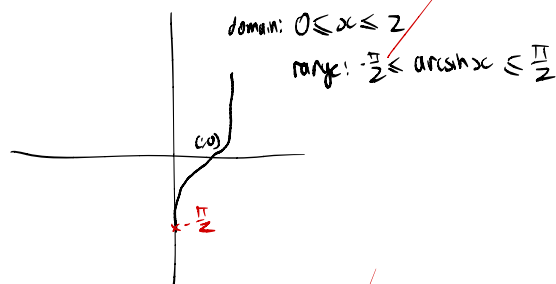
$\arccos(\tan \frac{\pi}{4}) = \arccos 1 = 0$ ✓

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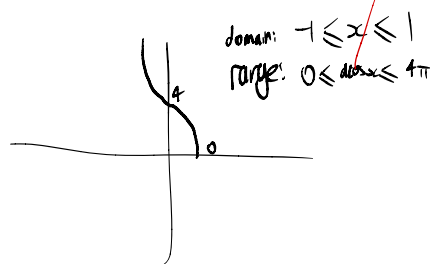
	domain	range
sinh	$-1 \leq x \leq 1$	$-\frac{\pi}{2} \leq \sinh x \leq \frac{\pi}{2}$
acos	$-1 \leq x \leq 1$	$0 \leq \arccos x \leq \pi$
atan	$-\infty \leq x \leq \infty$	$-\frac{\pi}{2} < \arctan x < \frac{\pi}{2}$

$x \in \mathbb{R}$

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Exam Qs

1 a) $-\arctan(\frac{1}{2}x - \frac{1}{3}\pi) = 0$ at A
 $\frac{1}{2}x - \frac{1}{3}\pi = 0$
 $x = \frac{2}{3}\pi$ $A = (\frac{2}{3}\pi, 0)$

At B, $x = 0$

$\therefore B_y = -\arctan(-\frac{1}{3}\pi)$
 $B = (0, 0.808)$

6 I translate by $(\frac{1}{2}\pi, 0)$

II stretch by scale factor 2 parallel to x direction

III reflect in y axis